ECON 1020: Principles of Microeconomics

Homework 5: Chapters 8–9

The Costs of Taxation; International Trade

Due on Wednesday, October 28 at 12:20 PM

Submissions are accepted either online through NTU COOL or as a hard copy at your recitation session.

Part A: The Costs of Taxation

1. A hypothetical competitive market for takeaway meals has demand $Q_D = 180 - 2P_B$ and supply $Q_S = 2P_S - 60$. Prices are NT$ per meal; quantities are meals per day. There are no externalities.

   (a) Find the no-tax equilibrium price and quantity, consumer surplus, and producer surplus.

   (b) A tax of NT$20 per meal is collected from sellers. Find P_B, P_S, and the quantity traded. Calculate consumer surplus, producer surplus, tax revenue, total surplus, and deadweight loss.

   (c) Draw the demand and supply curves. Label the no-tax equilibrium, post-tax prices and quantity, the revenue rectangle, and deadweight-loss triangle.

2. Continue with the meal market in Question 1. Let t be the tax per meal, with $0 \le t \le 60$.

   (a) Derive quantity traded as a function of t. Complete the table below. All revenue and deadweight-loss figures are NT$ per day.

Tax t	Quantity	Tax revenue	Deadweight loss
0			
10			
20			
30			
40			
60			

   (b) Sketch tax revenue as a function of t. Identify the tax that maximizes revenue and explain why revenue eventually falls. You may use algebra or symmetry; calculus is not needed.

   (c) Compare deadweight loss at taxes of NT$10 and NT$20. Explain why it changes by a different proportion from the tax. Would revenue maximization alone identify the best tax policy? Explain.

3. Explain the welfare logic in each situation. Use complete sentences.

   (a) In a hypothetical bicycle repair near NTU, a student values the repair at NT$1,500. The repairer's opportunity cost is NT$1,100. They agree on NT$1,300 before a tax is introduced. A tax of NT$500 is then imposed on the transaction. Find consumer and producer surplus before the tax. Will the repair occur after the tax? Find revenue and deadweight loss.

   (b) Two markets share the same no-tax price and quantity and the same supply curve, which is linear and upward-sloping. Both demand curves are linear; demand is more price elastic at the initial equilibrium in A than B. Compare deadweight loss from the same small per-unit tax. Explain why a perfectly inelastic demand curve would be a limiting case.

   (c) A worker's weekly hours do not change after a labor tax increases. Does this observation establish that the tax has no effect on total labor supplied in the economy? Discuss at least two other margins of adjustment.

   (d) A tax reduces consumer surplus by NT$900 and producer surplus by NT$600, and raises NT$1,200 in revenue per day. Calculate deadweight loss. Explain why the revenue is not itself a social loss.

Part B: International Trade

4. A hypothetical small economy has notebook demand $Q_D = 100 - P$ and supply $Q_S = P - 20$. Price is NT$ per notebook, and quantities are notebooks per day. Markets are competitive, with no externalities or transport costs. The economy takes the world price as given.

   (a) Find the no-trade equilibrium and consumer and producer surplus.

   (b) The economy opens to trade at a world price of NT$40. Find domestic production, domestic consumption, and imports or exports. Draw a diagram showing both the no-trade equilibrium and the outcome with trade.

   (c) Calculate consumer surplus, producer surplus, and total surplus with trade. Identify the winners and losers and calculate the net gain.

   (d) Instead, suppose the world price is NT$80. Find production, consumption, and trade. Calculate the change in each group's surplus relative to no trade. Does every resident necessarily benefit? Explain.

5. Return to the world price of NT$40 in Question 4. The government imposes a tariff of NT$10 per imported notebook.

   (a) Find the domestic price, production, consumption, imports, and tariff revenue. Explain why the tariff is not multiplied by total domestic sales when calculating revenue.

   (b) Draw a diagram showing the world price, tariff-inclusive price, quantities, revenue rectangle, and both deadweight-loss triangles.

   (c) Calculate consumer surplus, producer surplus, and total surplus. Relative to free trade, find the losses from distorted production and consumption separately.

   (d) What is the smallest tariff that eliminates imports? If the tariff were NT$30, what would the domestic price and tariff revenue be? Explain.

6. Continue with the notebook market and world price of NT$40.

   (a) Replace the tariff with a quota of 20 imported notebooks per day. Find the domestic price and total quota rents. Compare the recipients of these rents if permits are given to domestic importers, auctioned by the government, or given to foreign exporters.

   (b) Demand rises to $Q_D = 120 - P$, while supply and the world price stay unchanged. Compare domestic prices and imports under the NT$10 tariff and the quota of 20. Why do initially equivalent policies now differ?

   (c) Write a short paragraph evaluating the claim, "A tariff creates jobs in the protected industry, so it must improve national welfare." Address consumer losses, producer gains, revenue, and adjustment costs.

   (d) Choose either the national-security or infant-industry argument for protection. State the argument, one condition that could support it, and one difficulty in using it to justify a tariff. Illustrate your answer with a Taiwan example, real or hypothetical.